

Karthik Ragulan

IIT Kharagpur

☎ +91-7760057180 ✉ karthikr90637@gmail.com 🌐 LinkedIn 🐙 github

EDUCATION

Degree	Institute	Board/University	CGPA/Percentage	Year
Dual Degree	Indian Institute of Technology Kharagpur	IIT	7.35	2027
AISSCE	Narayana Junior College, Bangalore	CBSE	91.0%	2022
AISSE	Amara Jyothi Public School, Bangalore	CBSE	89.4%	2020

SKILLS AND COMPETENCIES

Languages : Python | SQL | C++ | Javascript

AI/ML Frameworks : Pytorch | Tensorflow | Scikit-learn | LangChain | LangGraph | Google ADK | FastAPI | Pandas

AI Skills : LLM Fine-tuning | RAG Systems | Multi-Agent Orchestration | Prompt Engineering | MLOps | Graphs

Cloud & Infra : GCP | Cloud Spanner | Vertex AI | GKE | Cloud Run | BigQuery | Docker | Pub/Sub | MongoDB Atlas

Tools : Git | GitHub | VS Code | Jupyter Notebook | Google Colab | MySQL | Notion | PyCharm

INTERSHIP

AI Sales Intelligence Agent | Glance · InMobi

May 2026 – Jul 2026

- Built a **Multi Agent** system orchestrating 5 specialized tools over a **14M-row, 64GB Spanner** merchant dataset
- Designed an **Account Memory Hub** via interleaved **Spanner** tables linking deal stage and rep attribution.
- Implemented **RAG on PDFs, contracts, transcripts** via embeddings and **Vector Distance** semantic search
- Tagged chunks with **graph-ready entities** (brands, objections, competitors) for future knowledge-graph assembly
- Enforced **propose→confirm** writes via schema-validated tools and auditable **LangGraph** checkpointer
- Implemented brand **similarity scoring** and fixed Spanner index-defeating patterns, achieving a **30× query speedup**

COMPETITIONS

Causal Analysis RAG System | Inter IIT 14.0 | Silver Medal

Nov 2025 – Dec 2025

- Built a multi-agent RAG system across **5 FastAPI microservices**, deployed via **Docker Compose** for horizontal scaling
- Architected a 3-hop **LangGraph** retriever with LLM sufficiency checks, iteratively refining queries until evidence is complete
- Combined **ColBERT, Cohere, and Voyage** into a weighted ensemble reranker, surfacing the top-10 most relevant evidences
- Integrated **Zep Cloud's GraphRAG** memory, enriching follow-up queries with prior context to skip redundant retrieval calls
- Automated **Presidio**-based PII redaction across 11 entity types, with exponential-backoff retries and **TF-IDF fallback** rerank.

DeloConnect: AI Employee Wellness Platform | Deloitte OpenSoft 2025 | Gold Medal

Feb 2025 – May 2025

- Built a **LangGraph** fan-out/fan-in workflow across 5 agents (activity, leave, performance, rewards etc) into HR reports
- Designed a counseling chatbot with an **Agno/OpenAI** decision agent triggering topic changes, **HR escalation**, or session end
- Built a **Gemini** summarizer converting escalated chats into structured HR reports flagging concerns, sentiment, and action items
- Engineered an **Isolation Forest** ensemble over EMA-smoothed activity, leave, and **Vibemeter** signals to flag at-risk employees
- Built a **FastAPI + MongoDB Atlas** backend with JWT auth and **WebSocket** chat enforcing HR-employee **RBAC**

AI-Driven Crop Disease Prediction System | Smart India Hackathon 2024 | Finalist

Aug24 – Sep24

- Fine-tuned **InceptionV3 CNN** on **25K+** crop images, achieving **95%** test accuracy across **10+** disease categories
- Built a cross-platform React Native mobile app used by **1000+** farmers, offering multilingual diagnosis in **11 languages**
- Deployed backend with **FastAPI** and live **TensorFlow** inference, delivering multilingual output via TTS API
- Added real-time weather alert system, reducing crop failure risk by **30%+** through geo-targeted prevention guidance

PROJECTS

Vantage — B2B Marketing Intelligence Platform

Jun. 2026

- Architected a **Next.js** SaaS on **Aurora PostgreSQL**, modeling **500+** customer journeys across five ad channels
- Built **Shapley-value** attribution engine in Python; LinkedIn drove **34%** of conversions on **18%** of spend
- Wrote **PostgreSQL** window-function queries detecting per-channel CPC anomalies against **7-day** rolling baselines
- Deployed **Next.js** on **Vercel Edge Functions** backed by **AWS RDS**, authenticated via **NextAuth.js**